A basis-valued refinement of the DFJP factorization for separable BAP ranges

Abstract

We prove that if Y is separable and has the bounded approximation property, then every weakly compact operator $T : X \to Y$ factors through a reflexive Banach space with a Schauder basis. In particular, if Y has a Schauder basis, then every weakly compact operator into Y factors through a reflexive Banach space with a Schauder basis.

1 The statement and the two external ingredients

Theorem 1.1. *Let Y be a separable Banach space with the bounded approximation property. If $T : X \to Y$ is weakly compact, then T factors through a reflexive Banach space with a Schauder basis.*

Corollary 1.2. *If Y has a Schauder basis and $T : X \to Y$ is weakly compact, then T factors through a reflexive Banach space with a Schauder basis.*

We use the following two classical facts.

Fact 1.3 (DFJP interpolation)**.** *Let E be a Banach space and let $K \subset E$ be bounded, closed, absolutely convex and weakly compact. For $j \in \mathbb{Z}$ put*

$$U_j = 2^j K + 2^{-j} B_E,$$

let $|\cdot|_j$ be the Minkowski functional of U_j, and define

$$\Delta(K) = \left\{ z \in E : \ \|z\|^2_{\Delta(K)} = \sum_{j \in \mathbb{Z}} |z|_j^2 < \infty \right\}.$$

Then $\Delta(K)$ is reflexive, the inclusion $\Delta(K) \hookrightarrow E$ is bounded, and K is bounded as a subset of $\Delta(K)$.

Fact 1.4 (Johnson's reflexive basisification theorem)**.** *Every separable reflexive Banach space with the bounded approximation property is isomorphic to a complemented subspace of a reflexive Banach space with a Schauder basis.*

2 The finite-rank approximants

We first produce finite-rank approximants to T which converge on both the adjoint and the bidual side.

Lemma 2.1. *If Y is separable and $W \subset Y$ is weakly compact, then (W, w) is compact metrizable.*

Proof. Let $Y_0 = \overline{\mathrm{span}} W$. Then Y_0 is separable. Choose a norm-dense sequence (s_n) in the unit sphere of Y_0. For each n, choose $y_n^* \in B_{Y^*}$ such that $|y_n^*(s_n)| > 1/2$. The sequence (y_n^*) separates the points of Y_0: if $0 \neq y \in Y_0$, choose s_n with

$$\left\| s_n - \frac{y}{\|y\|} \right\| < \frac{1}{4}.$$

Then

$$\left| y_n^* \left(\frac{y}{\|y\|} \right) \right| > \frac{1}{4}.$$

Thus the map

$$W \longrightarrow \prod_{n=1}^{\infty} [-M, M], \qquad w \longmapsto (y_n^*(w))_{n=1}^{\infty},$$

where $M = \sup_{w \in W} \|w\|$, is a weakly continuous injection into a compact metric space. Since W is weakly compact and the target is Hausdorff, this injection is a homeomorphism onto its image. $\square$

Lemma 2.2. *Let Y be separable and let $T : X \to Y$ be weakly compact. Put*

$$K = (B_{Y^*}, \mathrm{w}^*)$$

and define

$$F : K \to X^*, \qquad F(y^*) = T^* y^*.$$

Then F, regarded as a norm-valued map into X^*, is Baire-one.

Proof. Let

$$W = \overline{T(B_X)}^{\mathrm{w}} \subset Y.$$

By weak compactness of T, the set W is weakly compact. By Lemma 2.1, (W, w) is compact metrizable. Choose a weakly dense sequence (w_j) in W.

For $y^*, z^* \in B_{Y^*}$,

$$\begin{aligned} \|T^* y^* - T^* z^*\| &= \sup_{x \in B_X} |(y^* - z^*)(Tx)| \\ &= \sup_{w \in W} |(y^* - z^*)(w)| \\ &= \sup_j |(y^* - z^*)(w_j)|. \end{aligned} \tag{1}$$

The second equality uses weak density, since $w \mapsto (y^* - z^*)(w)$ is weakly continuous on W.

The formula (1) shows first that $F(K)$ is norm separable: it embeds isometrically into the separable space $C(W, \mathrm{w})$ by restriction to W. It also shows that for each $z^* \in K$ and $r > 0$,

$$\{y^* \in K : \ \|F(y^*) - F(z^*)\| \le r\} = \bigcap_{j=1}^{\infty} \{y^* \in K : \ |(y^* - z^*)(w_j)| \le r\},$$

which is weak-star closed in K. Hence preimages of norm-open balls in the separable metric space $F(K)$ are F_σ subsets of K. By the Lebesgue–Hausdorff theorem for maps from a metric space into a separable metric space, F is Baire-one. $\square$

Lemma 2.3 (tail convexification)**.** *Let K be compact metrizable, let E be a Banach space, and let $f_n : K \to E$ be uniformly bounded norm-continuous maps. Suppose that*

$$f_n(t) \xrightarrow{\mathrm{w}} f(t) \qquad (t \in K),$$

and that $f : K \to E$ is norm-Baire-one. Then there are

$$g_k \in \operatorname{conv}\{f_n : n \geq k\}$$

such that

$$\|g_k(t) - f(t)\| \longrightarrow 0 \qquad (t \in K).$$

Proof. Since f is bounded and Baire-one, choose uniformly bounded continuous maps $u_n : K \to E$ such that $u_n(t) \to f(t)$ in norm for every $t \in K$. Put $h_n = f_n - u_n$.

Let

$$L = K \times (B_{E^*}, \mathrm{w}^*).$$

For each n define

$$\widehat{h}_n(t, e^*) = e^*(h_n(t)) \qquad ((t, e^*) \in L).$$

Then $\widehat{h}_n \in C(L)$, the sequence $(\widehat{h}_n)$ is uniformly bounded, and $\widehat{h}_n \to 0$ pointwise on L. Hence $\widehat{h}_n \to 0$ weakly in $C(L)$, by dominated convergence against regular Borel measures on L.

By Mazur's theorem, for every k there is a finite convex combination

$$\widehat{v}_k = \sum_{n \geq k} a_{k,n} \widehat{h}_n$$

such that

$$\|\widehat{v}_k\|_{C(L)} \leq \frac{1}{k}.$$

Set

$$g_k = \sum_{n \geq k} a_{k,n} f_n, \qquad v_k = \sum_{n \geq k} a_{k,n} u_n.$$

For each $t \in K$,

$$\|g_k(t) - v_k(t)\| = \sup_{e^* \in B_{E^*}} \left| \sum_{n \geq k} a_{k,n} e^*(h_n(t)) \right| \leq \frac{1}{k}.$$

Also $v_k(t) \to f(t)$ for each fixed t, because $u_n(t) \to f(t)$ and the coefficients of v_k are supported in the tail $\{n : n \geq k\}$. Thus $g_k(t) \to f(t)$ in norm. $\square$

Proposition 2.4. *Let Y be separable with the bounded approximation property, and let $T : X \to Y$ be weakly compact. Then there are finite-rank operators $A_k : X \to Y$ such that*

$$A_k^* y^* \longrightarrow T^* y^* \qquad (y^* \in Y^*) \tag{1}$$

and

$$A_k^{**} x^{**} \longrightarrow T^{**} x^{**} \qquad (x^{**} \in X^{**}) \tag{2}$$

in norm.

Proof. Since Y is separable and has the bounded approximation property, there are uniformly bounded finite-rank operators $P_n : Y \to Y$ such that

$$P_n y \longrightarrow y \qquad (y \in Y).$$

Put $K = (B_{Y^*}, \mathrm{w}^*)$. Since Y is separable, K is compact metrizable. Define

$$F_n : K \to X^*, \qquad F_n(y^*) = T^* P_n^* y^*,$$

and

$$F : K \to X^*, \qquad F(y^*) = T^* y^*.$$

Each F_n is norm-continuous, because P_n has finite rank.

For $x^{**} \in X^{**}$ and $y^* \in K$,

$$\langle F_n(y^*) - F(y^*), x^{**} \rangle = \langle y^*, (P_n - I) T^{**} x^{**} \rangle.$$

Since T is weakly compact, $T^{**}x^{**} \in Y$ under the canonical embedding of Y into Y^{**}. Hence $P_n T^{**} x^{**} \to T^{**} x^{**}$ in norm. Therefore

$$F_n(y^*) \xrightarrow{\mathrm{w}} F(y^*) \qquad (y^* \in K).$$

By Lemma 2.2, F is norm-Baire-one. Applying Lemma 2.3, we get operators

$$S_k \in \operatorname{conv}\{P_n : n \geq k\}$$

such that

$$\|T^* S_k^* y^* - T^* y^*\| \longrightarrow 0 \qquad (y^* \in B_{Y^*}).$$

By homogeneity the same convergence holds for all $y^* \in Y^*$.

Set

$$A_k = S_k T.$$

Then each A_k is finite-rank, and (1) holds.

Finally, for $y \in Y$,

$$\|S_k y - y\| \leq \sup_{n \geq k} \|P_n y - y\| \longrightarrow 0.$$

Thus, for $x^{**} \in X^{**}$,

$$A_k^{**} x^{**} = S_k^{**} T^{**} x^{**} = S_k T^{**} x^{**} \longrightarrow T^{**} x^{**},$$

because $T^{**}x^{**} \in Y$. This proves (2). $\square$

3 The bridge theorem

The next proposition is the main structural step. It converts the approximants from Proposition 2.4 into a genuine factorization through a reflexive space with a basis.

Proposition 3.1 (bridge theorem)**.** *Let $T : X \to Y$ be a bounded operator. Suppose there are finite-rank operators $A_n : X \to Y$ such that*

$$A_n^* y^* \longrightarrow T^* y^* \qquad (y^* \in Y^*) \tag{3}$$

and

$$A_n^{**} x^{**} \longrightarrow T^{**} x^{**} \qquad (x^{**} \in X^{**}) \tag{4}$$

in norm. Then T factors through a reflexive Banach space with a Schauder basis.

Proof. By uniform boundedness, let

$$M = \sup_n \|A_n\| < \infty.$$

Condition (4), applied to canonical images of elements of X, implies $A_n x \to Tx$ in norm for every $x \in X$. Also (4) implies that T is weakly compact, since $A_n^{**} x^{**} \in Y$ and Y is norm closed in Y^{**}.

Let

$$E = c(Y) = \{(y_n) : y_n \text{ converges in norm in } Y\},$$

with the supremum norm. Define

$$R : X \to E, \qquad Rx = (A_n x)_{n=1}^{\infty},$$

and

$$L : E \to Y, \qquad L((y_n)) = \lim_n y_n.$$

Then

$$T = LR.$$

We first prove that R is weakly compact. Identify

$$c(Y) \cong c_0(Y) \oplus_\infty Y$$

by

$$(y_n) \longmapsto ((y_n - \lim_m y_m)_n, \lim_m y_m).$$

Under this identification,

$$Rx = \big((A_n - T)x\big)_n \oplus Tx.$$

Let $B : X \to c_0(Y)$ be given by

$$Bx = ((A_n - T)x)_n.$$

For $x^{**} \in X^{**}$ put

$$b(x^{**}) = \big((A_n^{**} - T^{**})x^{**}\big)_n.$$

By (4), $b(x^{**}) \in c_0(Y)$. If $(y_n^*) \in \ell_1(Y^*) = c_0(Y)^*$, then

$$\begin{aligned}
\langle B^{**}x^{**}, (y_n^*)\rangle &= \left\langle x^{**}, \sum_{n=1}^{\infty} (A_n - T)^* y_n^* \right\rangle \\
&= \sum_{n=1}^{\infty} \langle (A_n^{**} - T^{**})x^{**}, y_n^* \rangle \\
&= \langle b(x^{**}), (y_n^*) \rangle.
\end{aligned}$$

Thus $B^{**}x^{**} = b(x^{**}) \in c_0(Y)$. Since also $T^{**}x^{**} \in Y$, we have $R^{**}(X^{**}) \subset E$. Hence R is weakly compact.

Now define the freezing projections $Q_m : E \to E$ by

$$Q_m(y_1, y_2, \ldots) = (y_1, \ldots, y_m, y_m, y_m, \ldots).$$

Then

$$\|Q_m\| \le 1, \qquad Q_m Q_k = Q_{\min(m,k)}, \qquad Q_m z \to z \quad (z \in E).$$

Let

$$W = \overline{R(B_X)}^{\mathrm{w}} \subset E.$$

Since R is weakly compact, W is weakly compact.

We need the following shrinking estimate. Every $\Phi \in E^*$ has a representation

$$\Phi((z_n)) = \eta^*\Big(\lim_n z_n\Big) + \sum_{n=1}^{\infty} y_n^*(z_n), \tag{5}$$

where $\eta^* \in Y^*$ and $(y_n^*) \in \ell_1(Y^*)$. Indeed, this follows from the splitting $c(Y) = c_0(Y) \oplus Y$.

For $x \in X$,

$$(I - Q_m)Rx = (0, \ldots, 0, (A_{m+1} - A_m)x, (A_{m+2} - A_m)x, \ldots),$$

whose limit is $(T - A_m)x$. Applying (5) gives

$$R^*(I - Q_m^*)\Phi = (T^* - A_m^*)\eta^* + \sum_{n>m} (A_n^* - A_m^*)y_n^*.$$

Consequently,

$$\|R^*(I - Q_m^*)\Phi\| \le \|(T^* - A_m^*)\eta^*\| + 2M \sum_{n>m} \|y_n^*\|.$$

By (3),

$$\boxed{\|R^*(I - Q_m^*)\Phi\| \longrightarrow 0 \qquad (\Phi \in E^*)}. \tag{6}$$

Define

$$\mathcal{K} = \overline{\mathrm{aco}}\left(\bigcup_{m=1}^{\infty} Q_m W\right), \tag{7}$$

where the closure is the norm closure and aco denotes the absolutely convex hull. We claim that $\mathcal{K}$ is weakly compact.

It is enough to show that $\bigcup_m Q_m W$ is relatively weakly compact. Take a sequence $Q_{m_j} w_j$, where $w_j \in W$. Passing to a subsequence, either m_j is constant or $m_j \to \infty$. In the first case, weak compactness of $Q_m W$ gives a weakly convergent subsequence. In the second case, pass to a further subsequence so that $w_j \to w$ weakly in W. For $\Phi \in E^*$, (6) gives

$$\begin{aligned}\sup_{v \in W} |\Phi(Q_m v - v)| &= \sup_{x \in B_X} |\Phi(Q_m Rx - Rx)| \\ &= \|R^*(Q_m^* - I)\Phi\| \longrightarrow 0.\end{aligned}$$

Hence $\Phi(Q_{m_j} w_j - w_j) \to 0$, and so $Q_{m_j} w_j \to w$ weakly. Thus $\bigcup_m Q_m W$ is relatively weakly compact. By Krein's theorem, the closed absolutely convex hull $\mathcal{K}$ is weakly compact.

Moreover,

$$W \subset \mathcal{K}, \tag{8}$$

because $Q_m w \to w$ in norm for every $w \in W$, and

$$Q_m \mathcal{K} \subset \mathcal{K} \tag{9}$$

because $Q_m Q_k = Q_{\min(m,k)}$.

Apply the DFJP construction, Fact 1.3, to the weakly compact set $\mathcal{K} \subset E$. Thus, for $j \in \mathbb{Z}$, put

$$U_j = 2^j \mathcal{K} + 2^{-j} B_E,$$

let $|\cdot|_j$ be the Minkowski functional of U_j, and set

$$Z = \left\{ z \in E : \ \|z\|_Z^2 = \sum_{j\in\mathbb{Z}} |z|_j^2 < \infty \right\}.$$

Then Z is reflexive, $Z \hookrightarrow E$ continuously, and $\mathcal{K}$ is bounded in Z. Since $R(B_X) \subset W \subset \mathcal{K}$, the operator $R : X \to E$ lifts to a bounded operator

$$\widetilde{R} : X \to Z.$$

Therefore

$$T = LJ\widetilde{R},$$

where $J : Z \hookrightarrow E$ is the inclusion.

It remains to arrange a basis. We first show that Z has the bounded approximation property. By (9) and $\|Q_m\| \le 1$, each Q_m maps U_j into U_j. Hence Q_m acts contractively on Z. Also, for fixed j and $z \in Z$,

$$|(I - Q_m)z|_j \le 2^j \|(I - Q_m)z\|_E \longrightarrow 0,$$

while

$$|(I - Q_m)z|_j \le |z|_j + |Q_m z|_j \le 2|z|_j.$$

Dominated convergence in the $\ell_2(\mathbb{Z})$-sum gives

$$\|Q_m z - z\|_Z \longrightarrow 0 \qquad (z \in Z). \tag{10}$$

We next show that $Q_m|_Z$ has finite rank. Let

$$H = \overline{\operatorname{span}}^E \mathcal{K}.$$

Then $Z \subset H$. Indeed, if $q : E \to E/H$ is the quotient map, then $q(\mathcal{K}) = 0$, and so

$$q(U_j) \subset 2^{-j} B_{E/H}.$$

Thus

$$|z|_j \ge 2^j \|qz\| \qquad (j \in \mathbb{Z}).$$

If $qz \neq 0$, then $\sum_{j\ge 0} |z|_j^2 = \infty$, hence $z \notin Z$. Therefore $Z \subset H$.

Fix m and put

$$F_m = \sum_{i=1}^{m} Q_i R(X).$$

Each $Q_i R(X)$ is finite-dimensional, since

$$Q_i R x = (A_1 x, \ldots, A_i x, A_i x, A_i x, \ldots)$$

and the $A_1, \ldots, A_i$ have finite-dimensional ranges. Thus F_m is finite-dimensional. If $w \in W$, then $Q_i w \in Q_i R(X)$, because $Q_i R(X)$ is finite-dimensional and hence weakly closed. Therefore, for $k \ge 1$,

$$Q_m Q_k w = Q_{\min(m,k)} w \in F_m.$$

By absolute convexity and norm closure, $Q_m \mathcal{K} \subset F_m$. Since $Z \subset H = \overline{\operatorname{span}}^E \mathcal{K}$, it follows that

$$Q_m Z \subset F_m.$$

Thus $Q_m|_Z$ is finite-rank.

Combining this finite-rank property with (10), we see that Z has the bounded approximation property. Also Z is separable, since $\bigcup_m Q_m Z$ is dense in Z and each $Q_m Z$ is finite-dimensional.

By Fact 1.4, Z is isomorphic to a complemented subspace of some reflexive Banach space V with a Schauder basis. Let $i : Z \to V$ be the embedding and let $P : V \to iZ$ be a bounded projection. Define

$$B : X \to V, \qquad B = i\widetilde{R},$$

and

$$C : V \to Y, \qquad C = (LJ)i^{-1}P.$$

Then $CB = T$. Hence T factors through the reflexive Banach space V with a Schauder basis. □

Proof of Theorem 1.1. Apply Proposition 2.4 to obtain finite-rank operators (A_n) satisfying (1) and (2). Then apply Proposition 3.1. □

Proof of Corollary 1.2. A Banach space with a Schauder basis is separable and has the bounded approximation property. Hence Theorem 1.1 applies. □

Remark 3.2. The proof does not use the basis of Y directly. It uses only a uniformly bounded sequence of finite-rank operators on Y converging strongly to the identity, which is available for separable spaces with the bounded approximation property. In the basis case, one may take the usual basis projections.

References

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